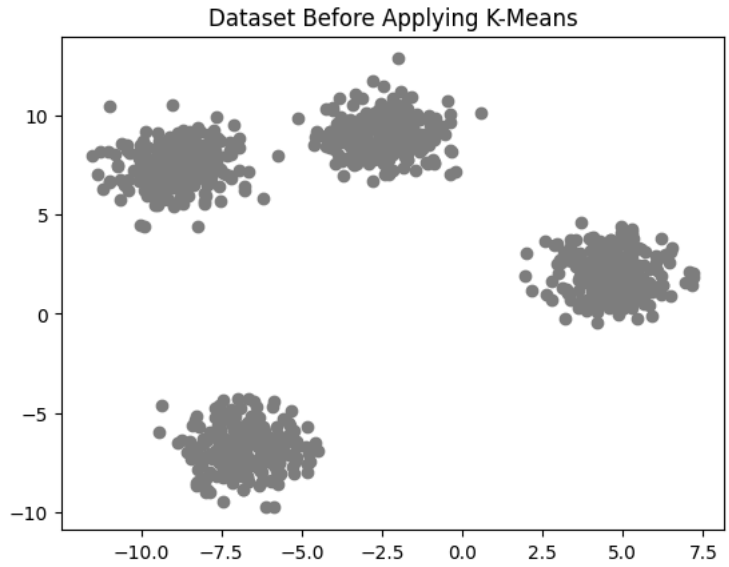
```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import make_blobs
from sklearn.cluster import KMeans
from skimage import data
```

```
from sklearn.datasets import make_blobs
import matplotlib.pyplot as plt

X, y = make_blobs(n_samples=1000, centers=4, random_state=42)
```

```
plt.scatter(X[:, 0], X[:, 1], color='gray')
plt.title('Dataset Before Applying K-Means')
plt.savefig('kmeans_data_before.png')
plt.show()
```

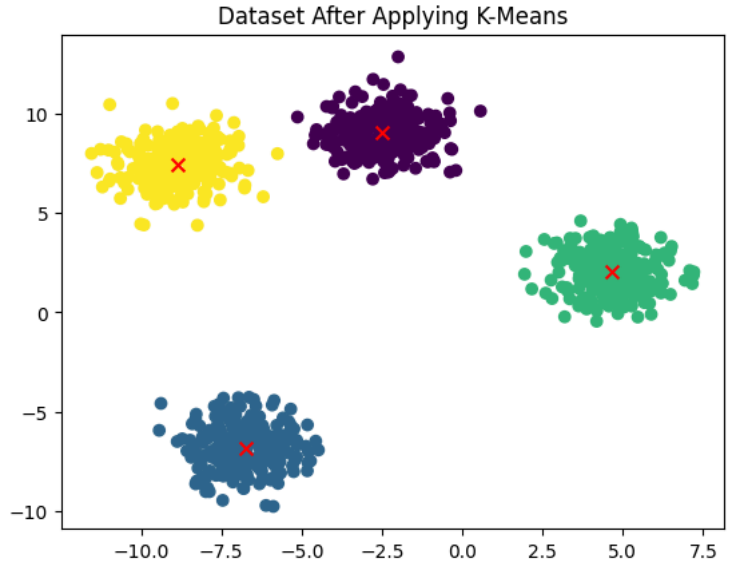



```
from sklearn.cluster import KMeans

kmeans = KMeans(n_clusters=4, random_state=42)
labels = kmeans.fit_predict(X)
```

```
plt.scatter(X[:, 0], X[:, 1], c=labels, cmap='viridis')
plt.scatter(kmeans.cluster_centers_[:, 0],
            kmeans.cluster_centers_[:, 1],
            color='red', marker='x', s=50)

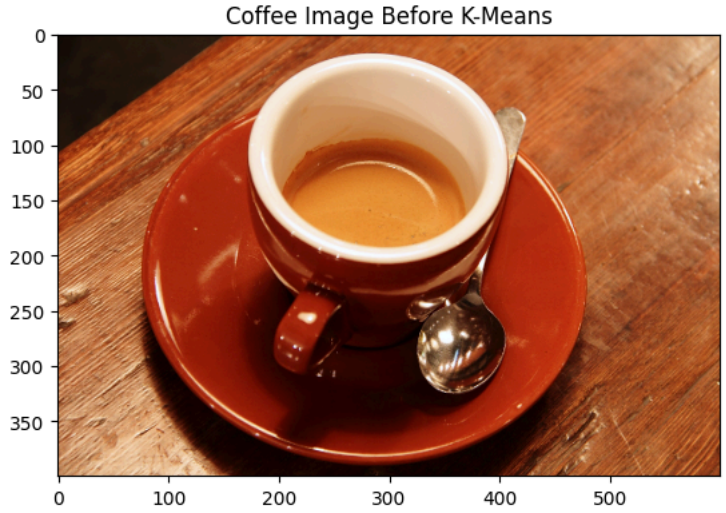
plt.title('Dataset After Applying K-Means')
plt.savefig('kmeans_data_after.png')
plt.show()
```



```
from skimage import data
import matplotlib.pyplot as plt
import numpy as np
from sklearn.cluster import KMeans

image = data.coffee()
```

```
plt.imshow(image)
plt.title('Coffee Image Before K-Means')
plt.savefig('kmeans_image_before.png')
plt.show()
```

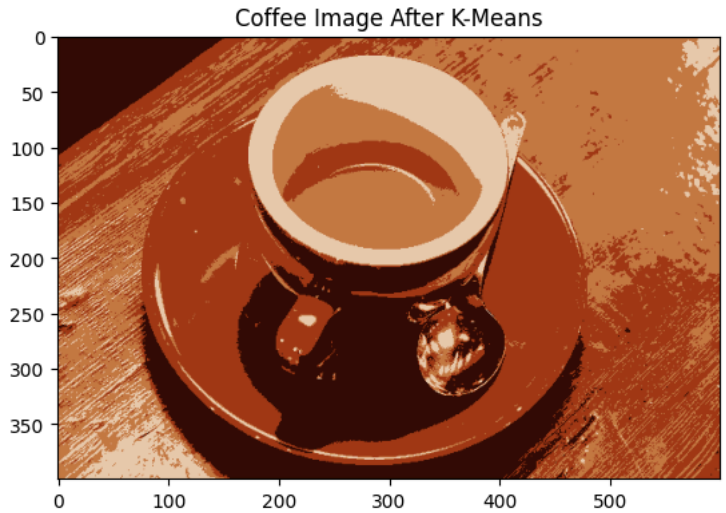



```
pixels = image.reshape(-1, 3)

kmeans = KMeans(n_clusters=4, random_state=42)
labels = kmeans.fit_predict(pixels)
```

```
new_pixels = kmeans.cluster_centers_[labels]
new_image = new_pixels.reshape(image.shape).astype(np.uint8)

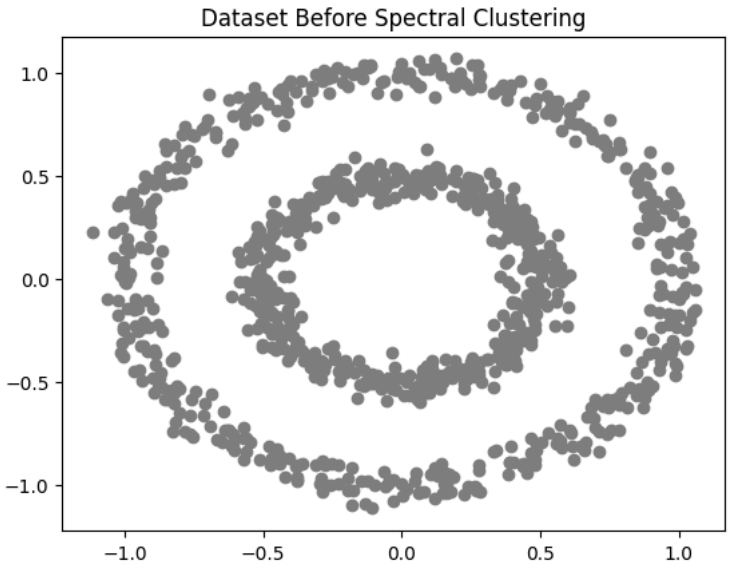
plt.imshow(new_image)
plt.title('Coffee Image After K-Means')
plt.savefig('kmeans_image_after.png')
plt.show()
```




```
from sklearn.datasets import make_circles
from sklearn.cluster import SpectralClustering
from sklearn.cluster import spectral_clustering
from sklearn.feature_extraction import image as sk_image
from skimage import data
from skimage.transform import rescale
from skimage.filters import gaussian
```

```
X, y = make_circles(n_samples=1000, factor=0.5,
                    noise=0.05, random_state=42)
```

```
plt.scatter(X[:, 0], X[:, 1], color='gray')
plt.title('Dataset Before Spectral Clustering')
plt.savefig('spectral_data_before.png')
plt.show()
```

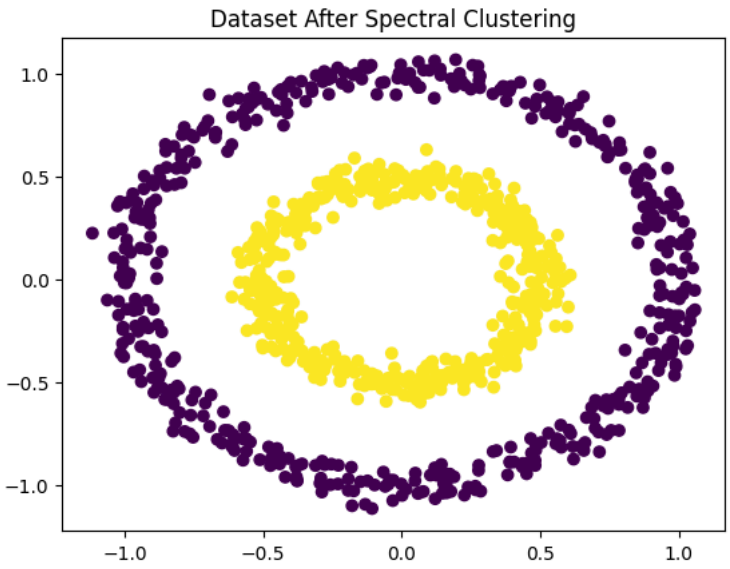



```
model = SpectralClustering(n_clusters=2,
                           affinity='nearest_neighbors',
                           random_state=42)

labels = model.fit_predict(X)
```

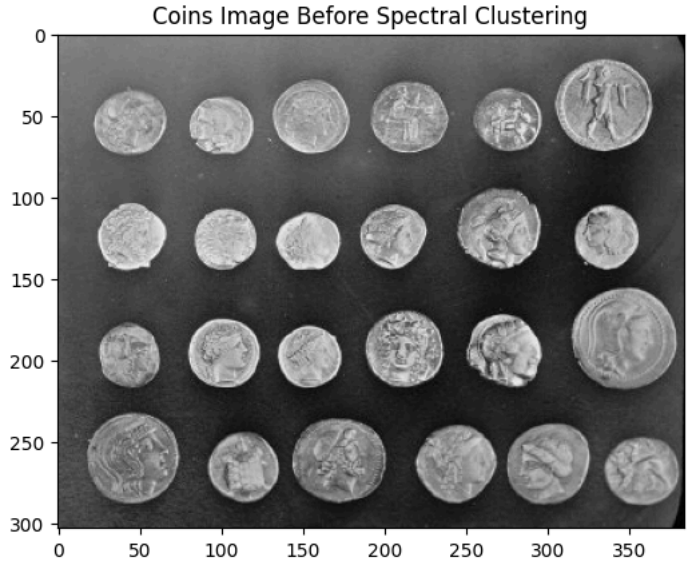
```
/usr/local/lib/python3.13/dist-packages/sklearn/manifold/_spectral_embedding.py:329: UserWarning: Graph is not fully connect
  warnings.warn(
```

```
plt.scatter(X[:, 0], X[:, 1], c=labels, cmap='viridis')
plt.title('Dataset After Spectral Clustering')
plt.savefig('spectral_data_after.png')
plt.show()
```




```
image = data.coins()
```

```
plt.imshow(image, cmap='gray')
plt.title('Coins Image Before Spectral Clustering')
plt.savefig('spectral_image_before.png')
plt.show()
```




```
smooth_image = gaussian(image, sigma=2)
small_image = rescale(smooth_image, 0.2, anti_aliasing=True)
```

```
graph = sk_image.img_to_graph(small_image)

graph.data = np.exp(
    -10 * graph.data / graph.data.std()
)

labels = spectral_clustering(
    graph,
    n_clusters=25,
    assign_labels='discretize',
    random_state=42
)

labels = labels.reshape(small_image.shape)
```

```
plt.figure(figsize=(10, 6))
plt.imshow(labels, cmap='nipy_spectral', interpolation='nearest')
plt.title('Coins Image After Spectral Clustering')
plt.axis('off')
plt.savefig('spectral_image_after.png', bbox_inches='tight')
plt.show()
```

